

KNOWLEDGE

Identify the choice that best completes the statement or answers the question by writing your answer on the line given. Show your reasons for your choice next to the question. Each question is worth 2 marks.

- a 1. The position function of an object moving horizontally along a straight line as a function of time is $s(t) = 8t^3 - 4t^2$ in metres, at time, t , in seconds. Determine the acceleration of the object at $t = 4$ s.

- a. 184 m/s²
b. 248 m/s²
c. 56 m/s²
d. 352 m/s²

Work/Explanation:

$$v(t) = 24t^2 - 8t$$

$$a(t) = 48t - 8$$

$$a(4) = 48(4) - 8 = 184 \text{ m/s}^2$$

- C 2. What is the second derivative of $f(x) = \frac{5}{3}x^3 - x^{\frac{3}{2}} + 6$?

- a. $f''(x) = 5x^2 - \frac{3}{2}\sqrt{x}$
b. $f''(x) = 10x + \frac{3}{4}\sqrt{x} + 6$
c. $f''(x) = 10x + \frac{3}{4\sqrt{x}}$
d. $f''(x) = 5x^2 - \frac{3}{2}\sqrt{x} + 6x$

Work/Explanation:

$$f'(x) = 5x^2 - \frac{3}{2}x^{\frac{1}{2}}$$

$$f''(x) = 10x - \frac{3}{4}x^{-\frac{1}{2}}$$

$$= 10x - \frac{3}{4\sqrt{x}}$$

- C 3. A basketball is thrown vertically upward for the tip off. Its position function $s(t)$, in metres, is $s(t) = -5t^2 + 4t + 2$. What is the maximum height the ball reaches if the players do not touch the ball until it begins to travel downward?

- a. 2.8 m
b. 0.8 m
c. 0.4 m
d. 5.6 m

Work/Explanation:

$$v(t) = -10t + 4$$

$$t = \frac{-4}{-10} = \frac{2}{5} = 0.4 \text{ m}$$

- b 4. Determine the absolute maximum value of the function $f(x) = -(4x - 2)^2 + 5$ on the interval $[-1, 2]$.

- a. 5
b. 0.5
c. -31
d. 1

Work/Explanation:

$$f'(x) = -2(4x - 2)(4)$$

$$0 = -32x + 16$$

$$x = 0.5$$

$$f(-1) = -31$$

$$f(0.5) = 5$$

$$f(2) = -31$$

5. Determine the equation of the tangent of the function, $x^2 + \frac{y^2}{4} = 1$ at $x = \frac{1}{2}$. Provide exact values.

$$2x + \frac{2}{4}y \frac{dy}{dx} = 0$$

$$\frac{1}{2}y \frac{dy}{dx} = -2x$$

$$\frac{dy}{dx} = \frac{-4x}{y} \rightarrow x = \frac{1}{2} \rightarrow \frac{-2}{\sqrt{3}}$$

$$\hookrightarrow x = \frac{1}{2} \rightarrow \frac{-2}{\sqrt{3}} = \frac{2}{\sqrt{3}}$$

relation

$$x + x = \frac{1}{2} \rightarrow \left(\frac{1}{2}\right)^2 + \frac{y^2}{4} = 1$$

$$\frac{1}{4} + \frac{y^2}{4} = 1$$

$$1 + y^2 = 4$$

$$y = \pm\sqrt{3}$$

tangents

$$y = \frac{2}{\sqrt{3}}x + b$$

$$\sqrt{3} = \frac{-1}{\sqrt{3}} + b$$

$$\frac{3}{\sqrt{3}} + \frac{1}{\sqrt{3}} = b \rightarrow b = \frac{4}{\sqrt{3}}$$

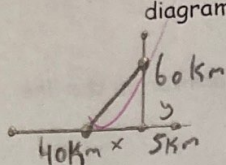
$$y = \frac{2}{\sqrt{3}}x + \frac{4}{\sqrt{3}}$$

$$-\sqrt{3} = \frac{1}{\sqrt{3}} + b$$

$$\frac{-3}{\sqrt{3}} - \frac{1}{\sqrt{3}} = -\frac{4}{\sqrt{3}} = b$$

$$y = \frac{2}{\sqrt{3}}x - \frac{4}{\sqrt{3}}$$

6. A bus leaves a bus station at 3pm and heads north at 60 km/h. At the same time, a car that is located 40km west of the bus station, is heading east at 45 km/h. When were the vehicles the closest together? (to the nearest minute). How far apart are they at this time in kilometres? Include a diagram in your solution.



$$0 \leq t$$

$\therefore$ The vehicles were closest together at 3:19 PM and were 32km apart at that point

$$D = d^2$$

$$d^2 = x^2 + y^2$$

$$d^2 = (40 - 45t)^2 + (60t)^2$$

$$D(t) = 1600 - 3600t + 2025t^2 + 3600t^2$$

$$D(t) = 5625t^2 - 3600t + 1600$$

$$D(t) = 11250t - 3600$$

$$t = 0.32 \text{ h} \rightarrow 60(0.32) = \text{about } 19 \text{ min}$$

$$D(0.32) = 5625(0.32)^2 - 3600(0.32) + 1600$$

$$= 576 - 1152 + 1600 = 1024$$

$$d^2 = 1024$$

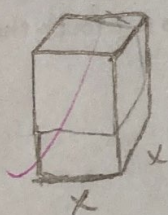
$$d = 32 \text{ km}$$

$$D(0) = 1600$$

$$d^2 = 1600$$

$$d = 40 \text{ km}$$

7. Thomas wants to design a square based prism with a volume of 12000 cm^3 . Determine the dimensions that will minimize the cost of material used, if the base and top costs $\$5/\text{cm}^2$, and the walls cost $\$2.50/\text{cm}^2$. The machinery available cannot fabricate material smaller than 4cm in length. Include let statements and a domain.



base is square,
 $4 \leq x$

$$V = x^2 h$$

$$12000 = x^2 h$$

$$h = \frac{12000}{x^2}$$

minimum x and h
is 4 (both are lengths of the fabric)

when $h = 4$

$$x^2 = 3000$$

$$x = 54.79 \text{ or } \sqrt{3000}$$

so $4 \leq x \leq \sqrt{3000}$
domain

Let $\left\{ \begin{array}{l} \text{Let } x = \text{length/width} \\ \text{Let } h = \text{height} \\ \text{Let } V = \text{volume} \\ \text{Let } C = \text{cost} \end{array} \right.$

$$h = \frac{12000}{(18.17)^2} = 36.35 \text{ cm}$$

$\therefore$ dimensions to minimize cost are $18.17 \text{ cm} \times 18.17 \text{ cm} \times 36.35 \text{ cm}$

$$C = 5(2x^2) + 2.5(4xh)$$

$$C = 10x^2 + 10xh$$

$$C(x) = 10x^2 + 10x \left(\frac{12000}{x^2} \right)$$

$$C(x) = 10x^2 + \frac{120000}{x}$$

$$C'(x) = 20x - \frac{120000}{x^2}$$

$$C'(x) = \frac{20x^3 - 120000}{x^2}$$

$$0 = 20x^3 - 120000$$

$$x^3 = 6000$$

$$x = 18.17 \text{ cm}$$

$$C(4) = 160 + 30000$$

$$= \$30160$$

$$C(18.17) = \$9905.78 \rightarrow \text{min}$$

$$C(\sqrt{3000}) = 30000 + 40$$

$$= \$30040$$

8. A particle moves along a horizontal line so that its position is given by the function $s(t) = t^3 - 8t^2 + 20t - 16$, $0 \leq t \leq 4$, where s is the position in meters and t is the time in seconds.

a) When does the particle change directions?

$$v(t) = 3t^2 - 16t + 20$$

$\therefore$ At 2s and 3.33s

1.5 (2)

$$0 = (t-2)(3t-10)$$

$$t = 2 \text{ s} \quad t = \frac{10}{3} \text{ s}$$

check it +ms.

b) When is the particle speeding up and slowing down?

$v \cdot a > 0$ is speeding up

$v \cdot a < 0$ is slowing down

$$a(t) = 6t - 16$$

$$t = \frac{16}{6} = 2.67 \text{ s}$$

Wib (+) when $t \in [0, 2]$

Test values

$$v(1) = (+)$$

$$v(4) = (+)$$

$$a(4) = (+)$$

$$v(3) = (-)$$

$$a(1) = (-)$$

v is (-) when $t \in (2, \frac{10}{3})$

a is (+) when $t \in (\frac{8}{3}, 4)$

a is (-) when $t \in [0, \frac{8}{3})$

s.u. when $t \in (2, \frac{8}{3}), (\frac{10}{3}, 4)$

s.d. when $t \in [0, 2), (\frac{8}{3}, \frac{10}{3})$

c) Create a position diagram to illustrate the particle's motion.

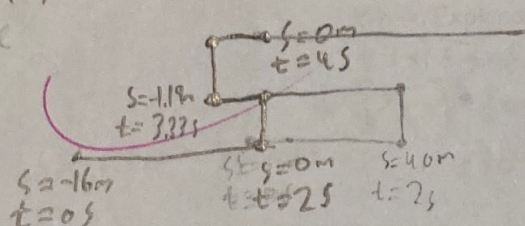
$$s(2) = 8 - 32 + 40 - 16 = 0 \text{ m}$$

$$s(\frac{10}{3}) = -1.19 \text{ m}$$

$$s(4) = 0 \text{ m}$$

$$s(6) = -16 \text{ m}$$

(2)



9. Let $y = \frac{2}{3}(x-a)^2 - \frac{1}{4}(x-b)^2$. Determine where y could have a local extrema, if a and b are constants.

$$(\frac{4}{3}x - \frac{1}{2}x) = \frac{8}{6}x - \frac{3}{6}x = \frac{5}{6}x$$

$$y = \frac{2}{3}(x^2 - 2ax + a^2) - \frac{1}{4}(x^2 - 2bx + b^2)$$

$$y' = \frac{2}{3}(2x - 2a) - \frac{1}{4}(2x - 2b)$$

$$y' = \frac{4}{3}x - \frac{4}{3}a - \frac{1}{2}x + \frac{1}{2}b$$

$$y' = \frac{5}{6}x - \frac{4}{3}a + \frac{1}{2}b$$

$$0 = \frac{5}{6}x - \frac{4}{3}a + \frac{1}{2}b$$

$$\frac{5}{6}x = \frac{4}{3}a - \frac{1}{2}b$$

$$5x = 8a - 3b$$

$$x = \frac{8a - 3b}{5} \rightarrow \text{here}$$

1.5 (2)

10. An object moves so that its velocity, v , is related to its position, s , according to

$v = 8\sqrt{b^2 + 4gs}$, where b and g are constants. Show that acceleration of the object is $128g$.

$$v = 8(b^2 + 4gs)^{\frac{1}{2}}$$

$$a = 4 \left(\frac{4g}{\sqrt{b^2 + 4gs}} \right) = \frac{4(4gv)}{\sqrt{b^2 + 4gs}} = \frac{16gv}{\sqrt{b^2 + 4gs}}$$

$$a = \frac{16g(8\sqrt{b^2 + 4gs})}{\sqrt{b^2 + 4gs}} = 16g(8) = 128g$$

$$\therefore a = 128g$$

(2)